EMG Sensors Manual

Electromyography and Motion Capture System
Noraxon Ultium EMG

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Date: December, 2025
Version: 1.0

Índice

1. Introduction and System Requirements

In this document, I present the procedure for the installation, assembly, and initial configuration of the **Noraxon Ultium EMG** system.

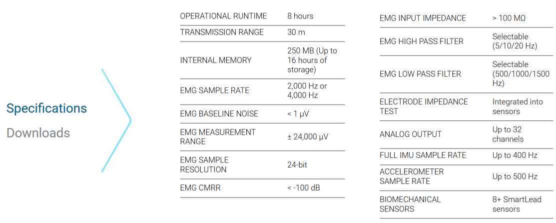
1.1. Operating System Compatibility

The Noraxon hardware and software ecosystem is natively designed to run on Microsoft Windows (10/11), meaning that, in principle, we cannot use Linux or Ubuntu.

1.2. Material

Figura 1: Material

1.3. Specifications



OPERATIONAL RUNTIME	8 hours
TRANSMISSION RANGE	30 m
INTERNAL MEMORY	250 MB (Up to 16 hours of storage)
EMG SAMPLE RATE	2,000 Hz or 4,000 Hz
EMG BASELINE NOISE	< 1 µV
EMG MEASUREMENT RANGE	± 24,000 µV
EMG SAMPLE RESOLUTION	24-bit
EMG CMRR	< -100 dB

EMG INPUT IMPEDANCE	> 100 MΩ
EMG HIGH PASS FILTER	Selectable (5/10/20 Hz)
EMG LOW PASS FILTER	Selectable (500/1000/1500 Hz)
ELECTRODE IMPEDANCE TEST	Integrated into sensors
ANALOG OUTPUT	Up to 32 channels
FULL IMU SAMPLE RATE	Up to 400 Hz
ACCELEROMETER SAMPLE RATE	Up to 500 Hz
BIOMECHANICAL SENSORS	8+ SmartLead sensors

Figura 2: Specifications

2. Software Installation (myoRESEARCH - MR3)

To get the system working and keep the drivers up to date, we need to install the **myoResearch (MR)** program on the computer.

Heads up: Don't worry about the USB Receiver drivers. The program installer itself includes them and installs them automatically.

We have two ways to do this. I personally used Option 2 (Web), but I explain both here:

2.1. Option 1: via USB

Inside the equipment case comes a USB drive with the software ready.

1. **Insert the USB:** Connect the flash drive included in the box to your PC.
2. **Install:** Open the installer file (it will be named something like `noraxon.mr.4.0.0`) and follow the instructions ("Next, Next...").
3. **Open:** When finished, you will see the icon on the desktop.
4. **Activate the License:**
   - Open the myoRESEARCH program.
   - You will see a warning that you are in trial mode. Click the **Activate** button.
   - Enter the **License ID** (the code) written on the USB itself or its paperwork and click OK.
   - If you have internet, click *Activate by Internet* and it activates instantly. If not, you would have to send an email to support.

2.2. Option 2: Web Download (The one I used)

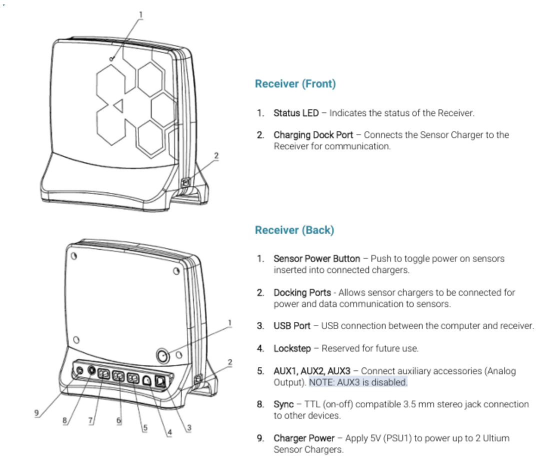
1. Access the official Noraxon download portal: https://www.noraxon.com/our-products/ultium-emg/#noraxon.
2. Download the package: **MR 4.2.22 Software** (or higher version available). The file in my folder is named noraxon.mr.4.2.22; otherwise, on the web, I would have to search here: https://www.noraxon.com/support-categories/software-downloads/ for the version MR 4.0.124 Software.
   Note: The file I have in my folder is named ***`noraxon.mr.4.2.22`***.
3. **Install and Activate:** Run the program and, just like in the previous option, when you open it, it will ask for the license. That is where we enter our code or contact the tutor if we don't have it.

Important Note: The MR3 installer automatically includes the USB drivers for the Ultium receiver. It is not necessary to install the *Noraxon USB Driver* separately.

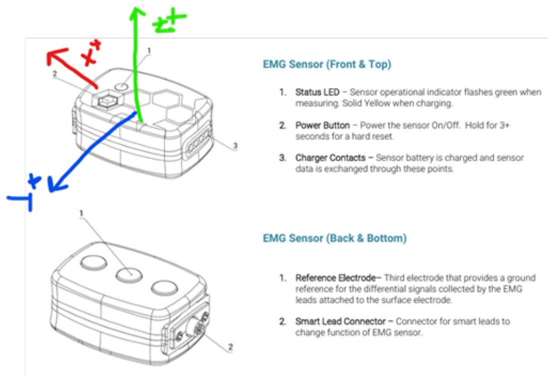
3. Hardware: Componentes y Montaje

3.1. Diagramas Técnicos

A continuación se detallan los componentes principales del sistema.



(a) Receptor Ultium (Puertos y Conexiones)



(b) Sensor EMG (Ejes IMU y Botón)

Figura 3: Diagramas de referencia de los componentes del sistema.

Installing the Hardware

1. Insert the USB cable (#CBL2) into the USB connector on the back of the Ultium Receiver (#880).
   Insert the opposite end of the USB cable into an available USB port on the computer.

2. Insert one EMG SmartLead (#842) into each EMG sensor (#810).

3. Insert the EMG sensors into the Sensor Charger (#873).
   Note: Sensor power button & LED must face the front of the Charger, i.e., toward the charger LEDs.

4. Connect the Receiver to the Sensor Charger using the ports on either side of the Receiver.

5. Insert the power supply (#PSU1) barrel connector into the associated receptacle on the back of the Ultium Receiver to charge sensors.
   Note: Sensor LED will show solid orange while charging and turn off when fully charged.

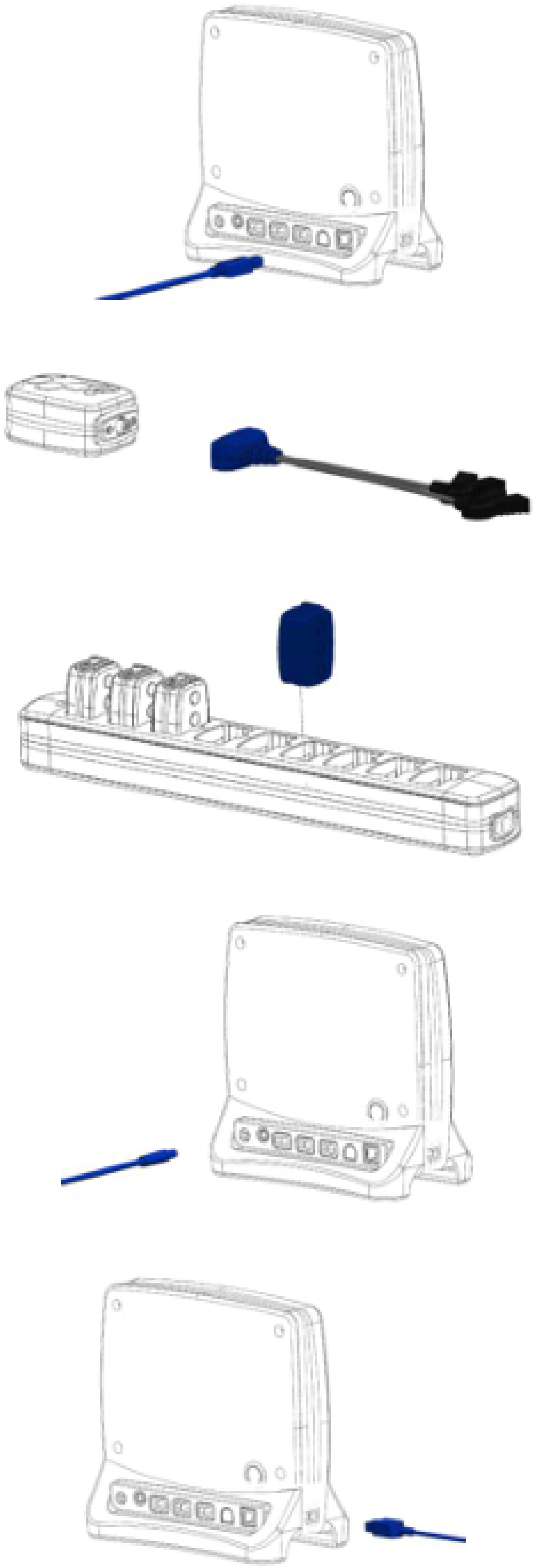


Figura 4: Guía visual del flujo de conexión del hardware.

3.2. Physical Status Verification

Once the assembly is complete, the system should look as follows. The connection of two simultaneous charging stations has been verified.

Figura 5: Final assembly: Receiver connected to the PC and two coupled charging stations with sensors inserted.

Interpretation of Status LEDs

- **Solid Amber/Orange Light:** The sensor is receiving power and charging the battery.
- **Light Off (in station):** Charge complete (100 %).
- **No Light (while others are charging):** Indicates poor physical contact. This is solved by adjusting the sensor in the slot or tightening the connection between stations.

(a) Charging detail (Amber LED) (b) General charging status

Figura 6: Visual verification of the sensors' charging status.

4. Configuring the Hardware

Before using the Ultium system, the device software settings must be configured to recognize the different components of the system. Follow the instructions below to update the receiver firmware, sensor firmware, and populate the sensor list to prepare for data collection.

Step 1: Open Hardware Setup Open the **myoRESEARCH (MR)** software. Click on the **Settings** icon located in the upper right-hand corner of the screen, then select **Hardware Setup**.

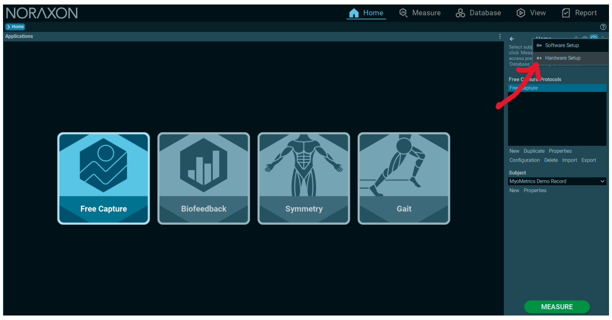


Figura 7: Opening Hardware Setup from the Home Screen.

Step 2: Connect and Insert Receiver After ensuring the Ultium Receiver is connected to a USB port on the computer:

1. Locate the **New Devices** area in the setup window.
2. Select the **Ultium Receiver** icon.
3. Click **Insert** to add it to your configuration.

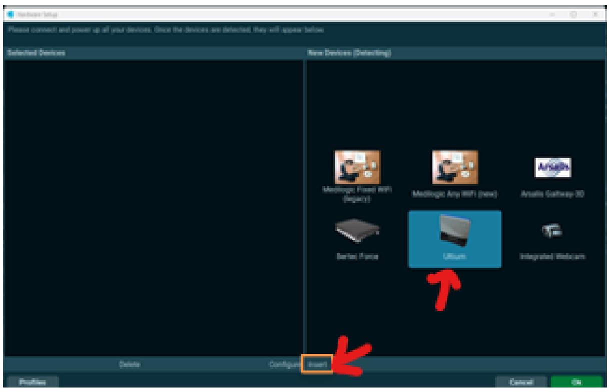


Figura 8: Inserting the Ultium Receiver in Hardware Setup.

Step 3: General Settings The Ultium hardware setup window will appear.

- **General Tab:** Select the desired settings for the system, such as the device name, sample rate, and cutoff filters.

- **Updates:** If firmware updates are available for the Receiver or Sensors, they will be indicated here.
- **Synchronization:** If using the synchronization system, check the box labeled **Use Noraxon myoSYNC**.

Note: Additional settings are available within the "Advanced" tab. If multiple Ultium systems are in use simultaneously, separate RF network assignments are required for each Receiver.

Step 4: Detect Sensors Ensure that all EMG or SmartLead Sensors are placed inside the Charger(s) and that the Charger is physically connected to the Ultium Receiver.

1. Click the button **Detect Sensors in Chargers**.
2. This will load all sensor serial numbers into the MR software.

Note: The order of the serial numbers in the list corresponds to the physical position of the sensors, starting from the front of the Charger (closest to the Charger LEDs).

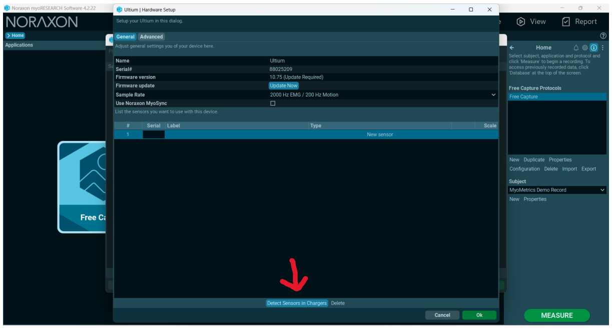


Figura 9: Detecting sensors in the charging station.

Step 5: Confirm Sensor List After detection is complete, a prompt will appear:

- Click **Replace** to overwrite any existing list with the currently detected sensors.
- Click **Add** to append the new sensors to an existing list.

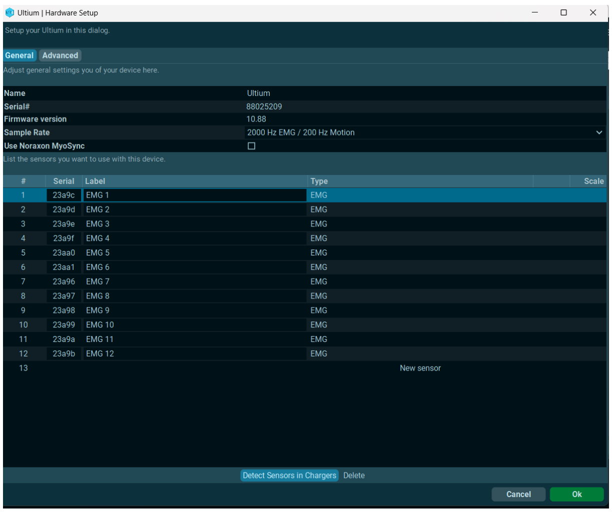


Figura 10: List of detected sensors populated in the setup.

Step 6: Save Click **OK** to save the hardware settings and close the setup window.

4.1. Configuring Ultium SmartLeads

Note: If you are only using surface EMG leads, you may skip this section and proceed to Section 5.

To configure SmartLeads (specialized sensors):

1. Connect the SmartLead to the Ultium sensor.
2. **Check LED Status:** The Ultium sensor LED will flash **purple** when a SmartLead is successfully connected. It will flash **orange** if disconnected.
3. Click **Detect Sensors in Charger** in the Hardware Setup. The system will list the SmartLead's specific serial number, rather than the generic Ultium sensor serial number.
4. Click **OK** to save the settings.

Note: If the software does not recognize the SmartLead, the sensors may require a firmware update.

5. Recording a Measurement

Step 1: Select Protocol On the Home screen:

1. Select or create a **Subject**.
2. Select an **Application**, then select or create a new **Protocol**.
3. Click **MEASURE** to proceed.

Step 2: Device Selection Insert the Ultium EMG device into the current configuration by dragging it from the **Available Devices** list to the **Devices in Your Configuration** area.

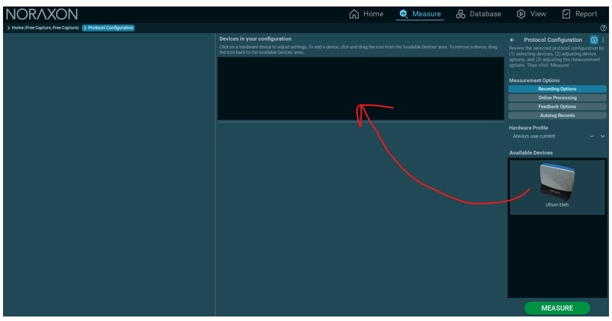


Figura 11: Adding the Ultium EMG device to the current configuration.

Step 3: Channel Assignment

1. Select the channels to include in the measurement by clicking their checkmarks.
2. Assign a label to each selected channel. You can do this by clicking the corresponding muscle on the **3D Muscle Map** or by entering the name manually in the list.
3. Click **MEASURE** to proceed.

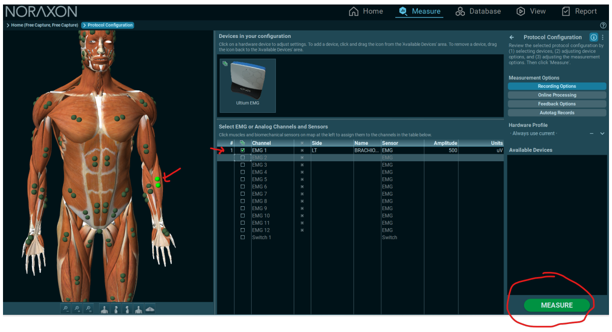


Figura 12: Assigning muscles and channels in the Protocol Configuration.

Step 4: Activation Click **ACTIVATE** to begin the live measurement stream.
Note: Any additional operations, such as measuring EMG impedance or creating a Maximum Voluntary Contraction (MVC) baseline, should be selected before clicking ACTIVATE.

Step 5: Record Data Verify the signal quality on the screen. Once the signal is stable, click **RECORD** to begin capturing data.

Figura 13: Live preview screen ready to record.

Step 6: Stop and Save (Internal Database)

1. Click **STOP** to end the recording.
2. Enter a name for the new record.
3. Select an action:
   - **Save & View:** To save and immediately begin processing/analyzing the record within MyoResearch.

- **Save & Measure Again:** To save and return to the measurement screen for another take.
- **Discard & Measure Again:** To delete the current trial and return to the measurement screen.

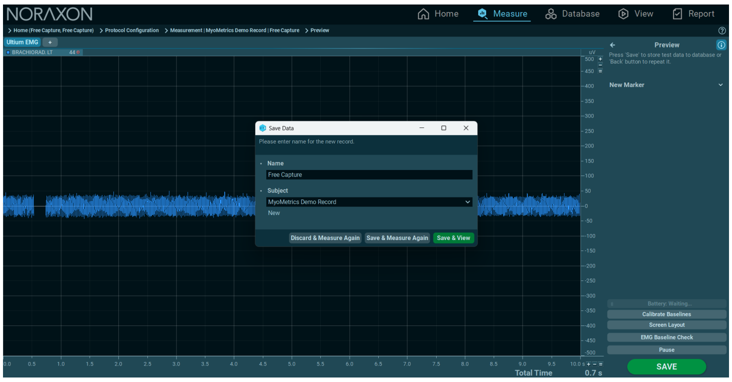


Figura 14: Saving the recorded data to the internal database.

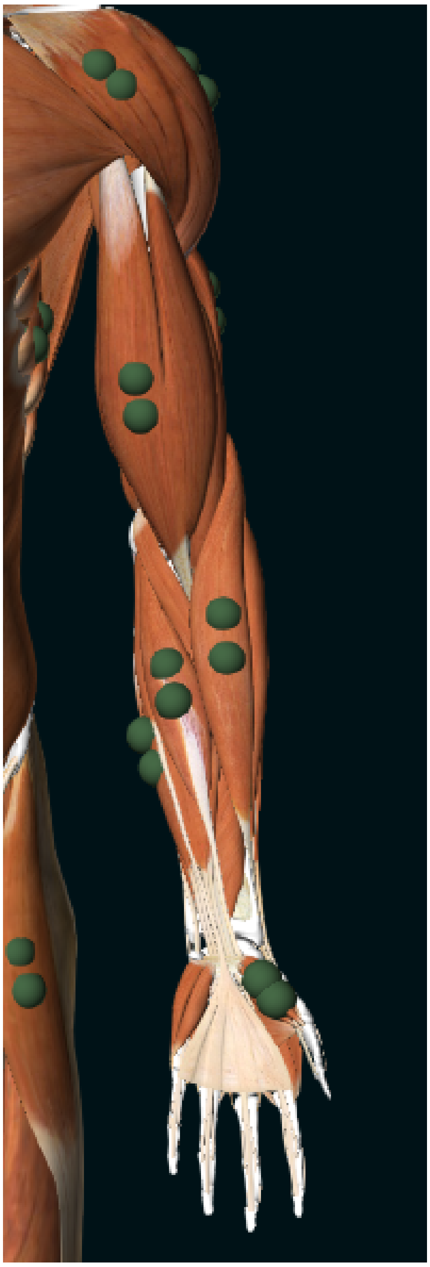
Step 7: Exporting Data for Python Processing To process the signal in the robotic control script, the data must be exported to an external CSV file using the regional format compatible with the script.

1. Navigate to the **Subject Database** tab (top menu).
2. Select the target record from the list.
3. Click the **Export** button in the right-hand tool panel.
4. Select the option **.Ex port Data to Single CSV Files"**.
5. **Configuration (Crucial for Python Script):**
   - **Format:** Select **CSV - Regional**.
   - *Note:* This setting uses a semicolon (`;`) as a separator and a comma (`,`) for decimals, matching the Python parameters `sep=';'` and `decimal=','`.
   - **Data Type:** Ensure the data is raw (unprocessed) to allow for custom filtering.
   - **Header:** Select "Short Header"to minimize metadata rows.
6. Click **OK** and save the file in your project folder (e.g., `/Pruebas/`).

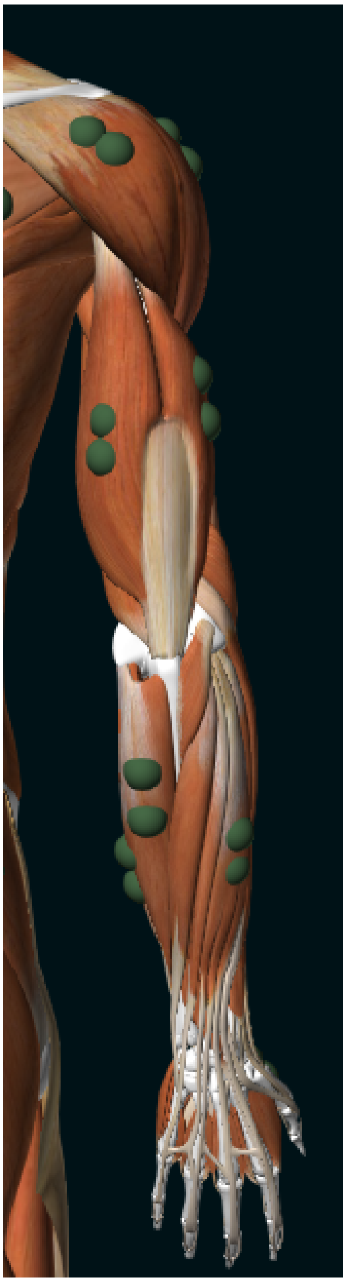
6. Preparation for Measurement

In the **MEASURE** screen, anatomical assignment is performed. Select a sensor channel and click on the corresponding muscle on the 3D map.

It is important to note that the system supports the possibility of using up to **5 sensors on each forearm** for detailed muscle analysis.



(a) Forearm Detail



(b) Measurement Interface

Figura 15: Measurement Interface: Assignment of sensors to muscle groups and visualization of active channels.

7. Activation of Inertial Measurement Unit (IMU) Sensors and Impedance Control

To integrate kinematics into this thesis project, it is essential to activate the Inertial Measurement Unit (IMU) modules embedded within the Noraxon Ultium sensors. This procedure is performed by accessing the `Hardware Setup` menu and navigating to the advanced sensor configuration.

Mandatory IMU Channels

The following options must be selected to ensure correct motion data acquisition:

- **ENABLE EMG IMU ACCEL**: Enables the high-range accelerometer (High-g). This sensor is essential for recording linear acceleration and body segment inclination, providing the foundation for kinematic analysis.
- **ENABLE EMG IMU GYRO AND MAG**: Activates the gyroscope and magnetometer. The gyroscope is critical for measuring angular velocity ($°/s$) and quantifying joint rotation (e.g., elbow flexion). Although the magnetometer is enabled simultaneously, its data may be disregarded during subsequent analysis if magnetic interference is detected in the robot's vicinity.

Quality Control

Additionally, it is recommended to enable the following option:

- **ALWAYS MEASURE EMG IMPEDANCE**: This function is crucial for quality control, as it verifies electrode impedance. Low impedance indicates optimal contact, ensuring the reliability of the EMG signal and minimizing baseline noise.

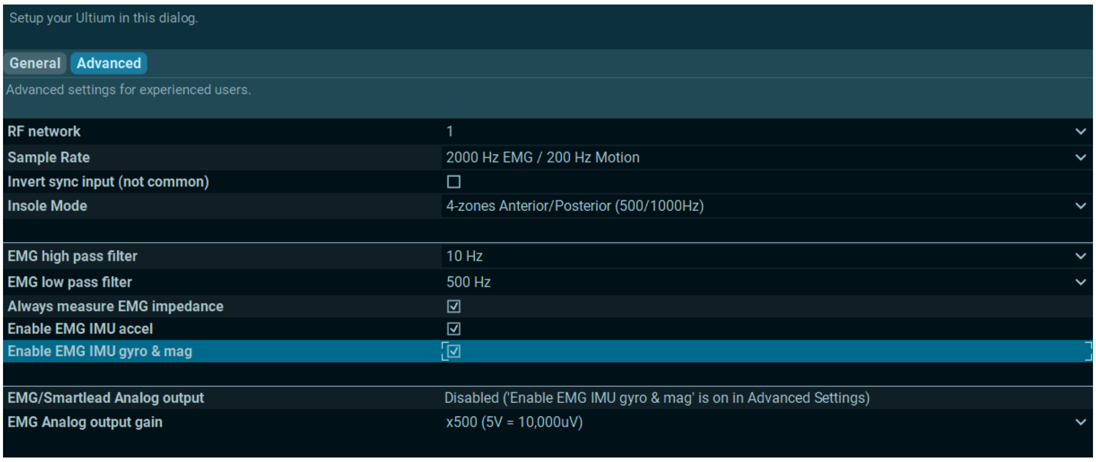


Figura 16: Advanced configuration of the Ultium sensor for enabling IMU channels (Accelerometer and Gyroscope) and impedance verification.

7.1. Requirement for Sensor Activation

Within the software, you must manually enable the accelerometer, gyroscope, and magnetometer for each individual sensor if you intend to utilize them. The following image demonstrates how to access these settings within the `Measure` section.

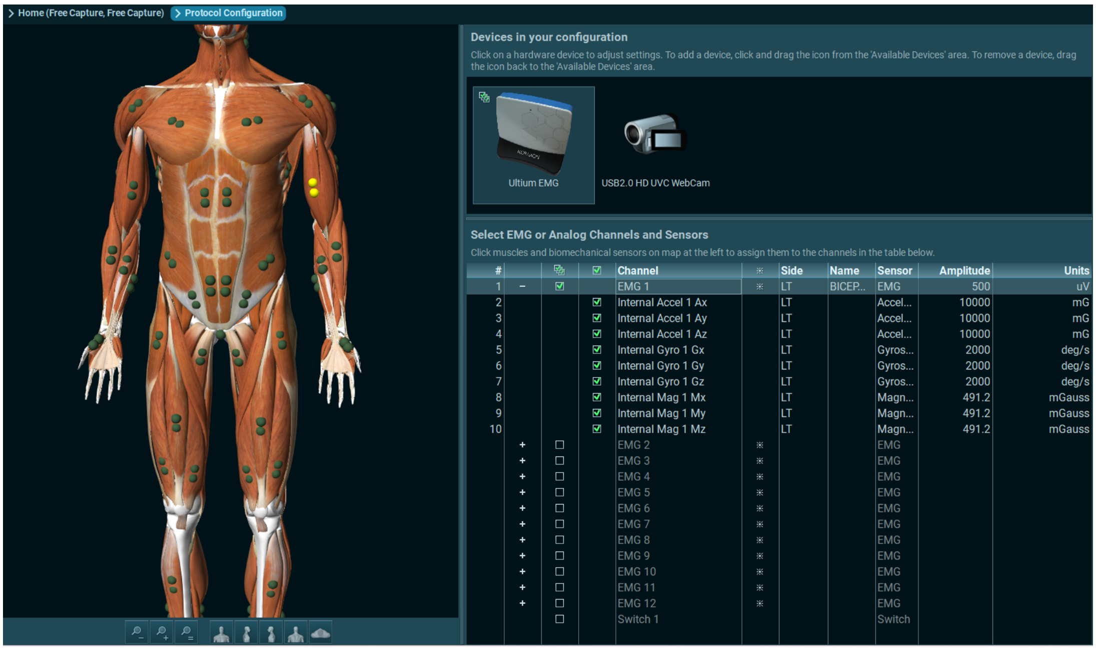


Figura 17: Manual activation of IMU channels within the myoRESEARCH Measure interface.

8. Protocol Creation and Saving

We can create our own protocol, so we can maintain constant sensor association and other configurations.

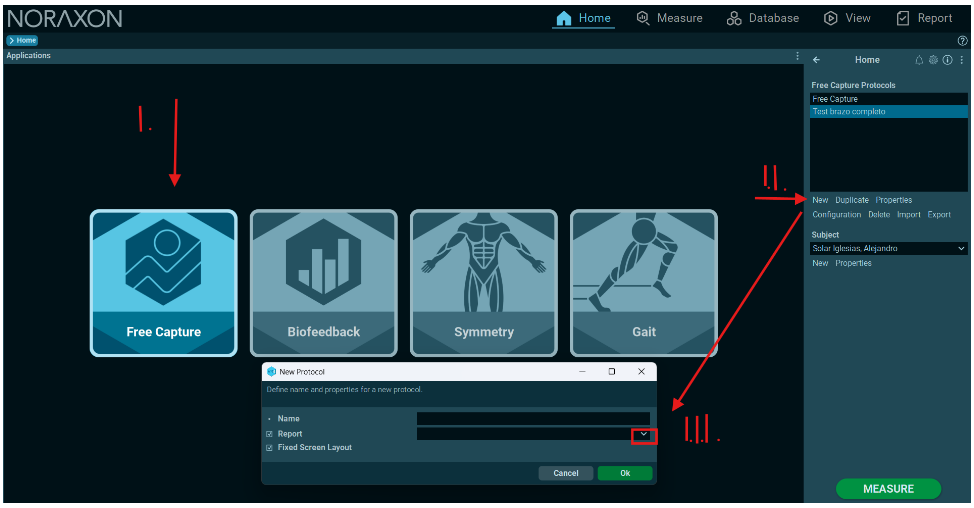


Figura 18: Protocol.

Choosing the final report is not necessary, but it can be done this way:

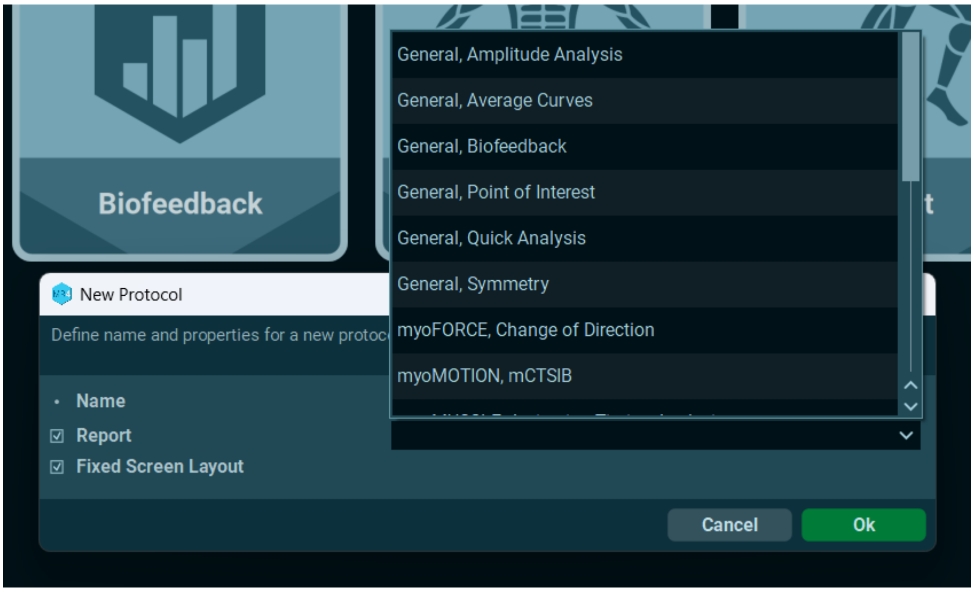


Figura 19: Report.

9. Real-time communication

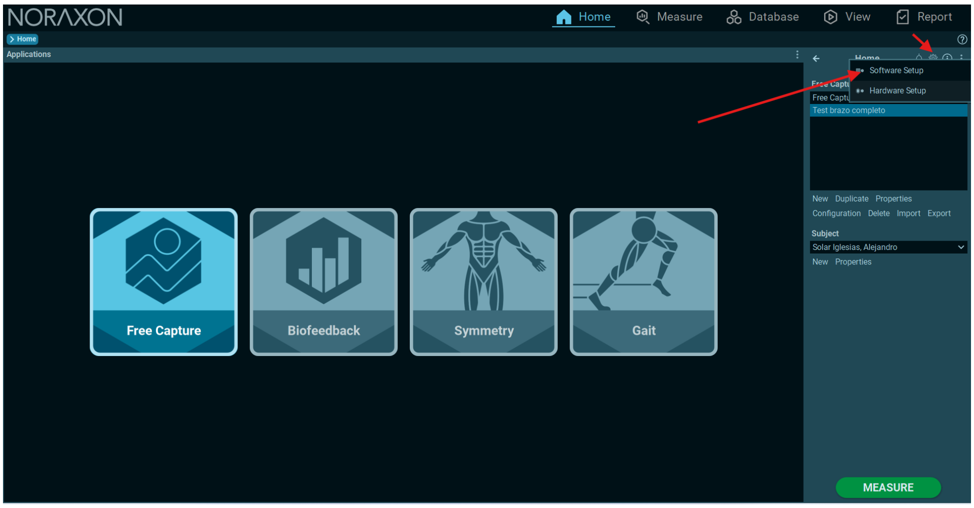


Figura 20: Select in Sotware.

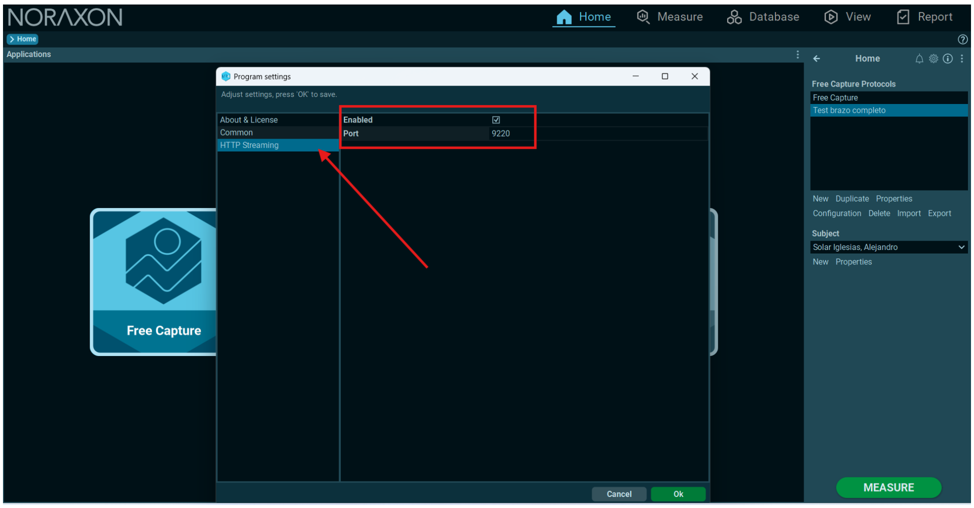


Figura 21: Activation

We can access it through a search engine and view the raw data signal, or via Python access here:

http://192.168.56.1:9220/samples

The published information is divided into indices corresponding to the different channels (EMG, acel x, acel y, acle z,..., mag z) of each signal.

- **192.168.56.1:**Current IP address of our PC where we have the Noraxón software
- **9220:**Port configured from the software.

10. Signal Processing

For signal processing, we can apply different filters and signal smoothing techniques, as shown in the following screenshots:

10.1. Real-Time Signal Processing

Below is an example of processing; we don't necessarily have to perform that on the signal.

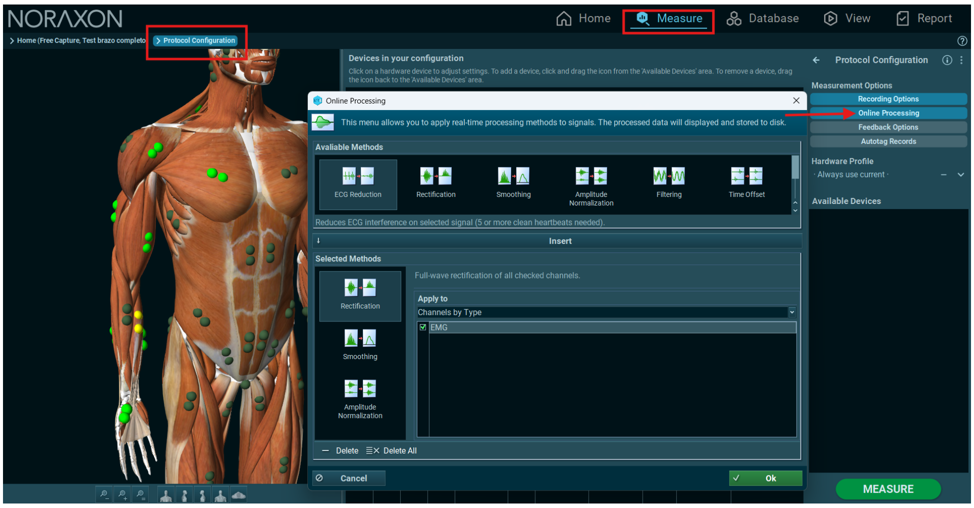


Figura 22: Real-time processing

10.2. Signal Recording Processing. Post-Processing

The first step is to enter the visualization of the recording that we want to process:

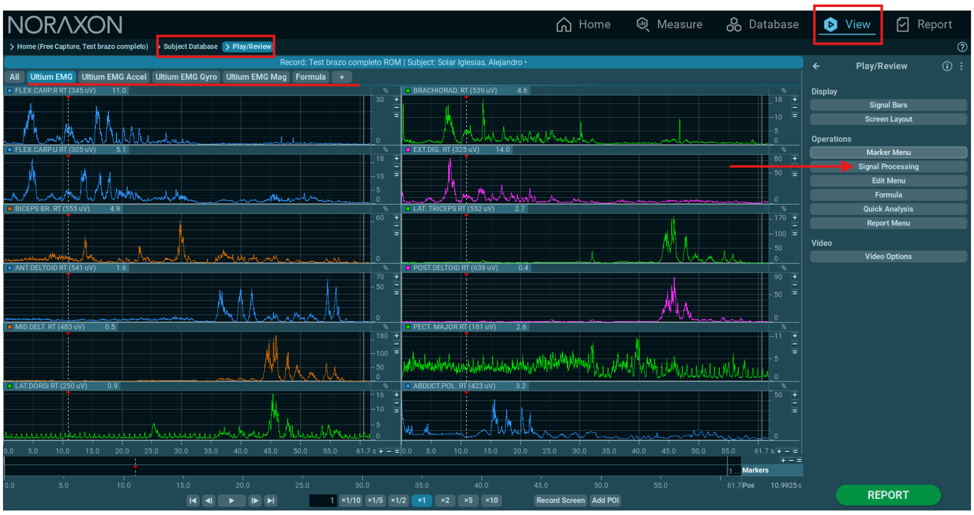


Figura 23: Recording Processing

We can also load or save a predefined processed run:

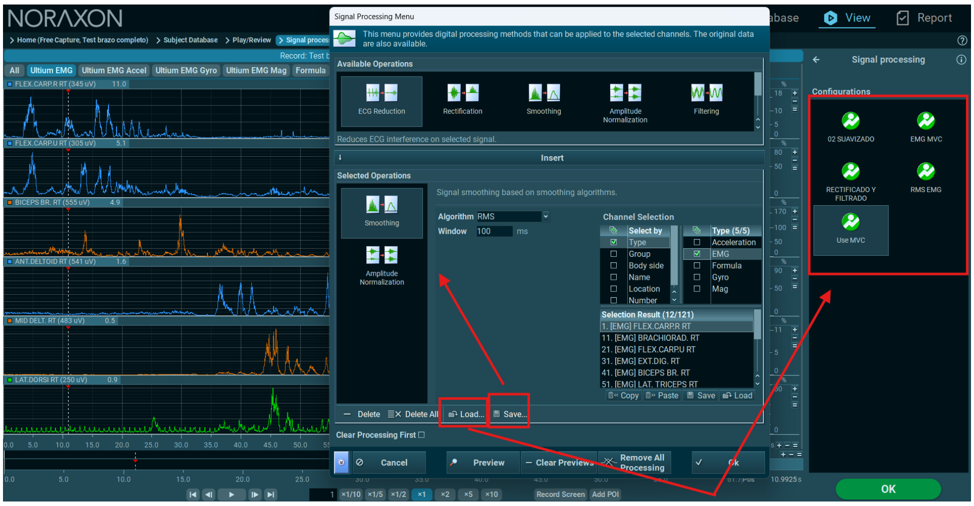


Figura 24: Saving and processing load

We can delete the processes applied to the signal:

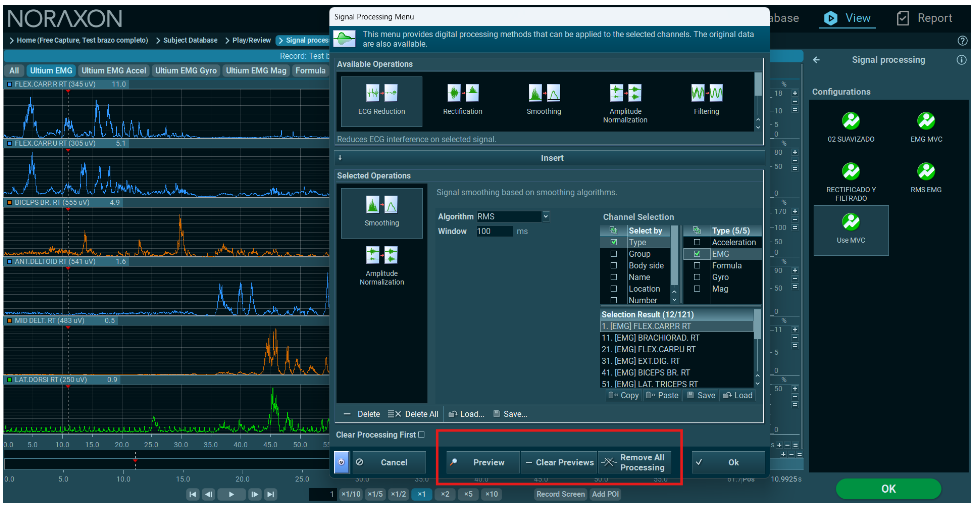


Figura 25: Processed cleaning

10.3. Signal Comparison

Example of comparison of raw and processed EMG signals:
Raw Signal:

Figura 26: Raw Signal

Example Processed Signal:

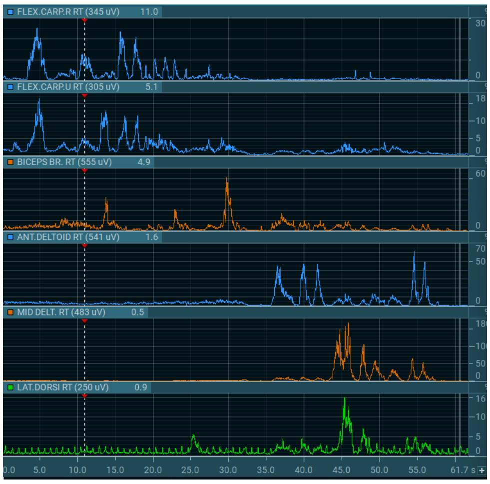


Figura 27: Processed Signal

11. Calibration Configuration

Below you can see the sensor configuration used, showing where each sensor is placed, as well as the name of the EMG and the muscle it is associated with:

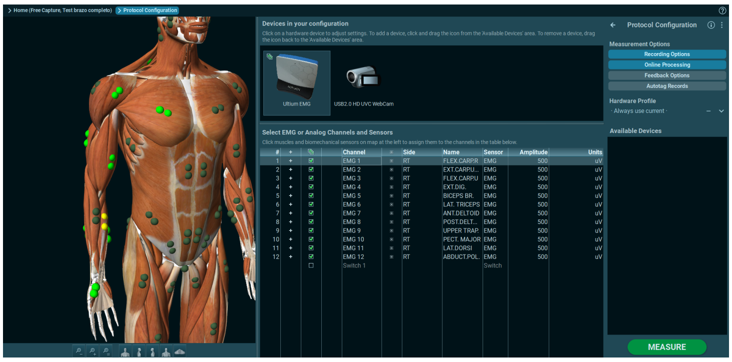


Figura 28: Configuration Emg

12. Reports

Below are examples of the reports that can be generated with Noraxon software, which allow visualization of the data obtained during testing:

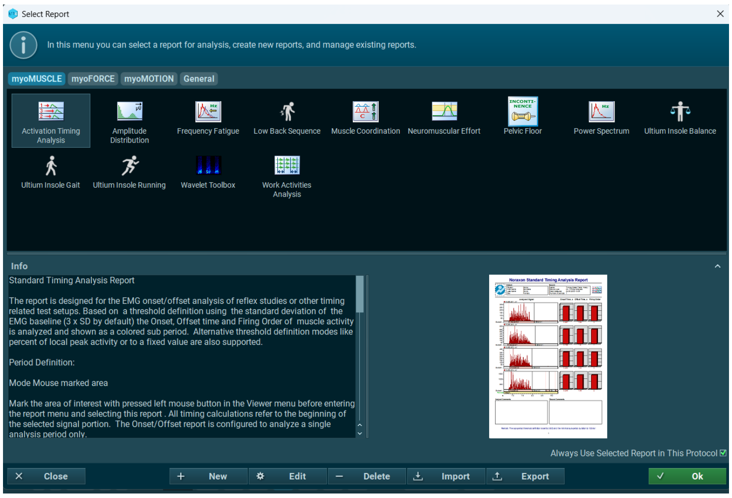


Figura 29: Examples of analysis reports in Noraxon software.